Contest Duration: 2019-01-06(Sun) 09:00 - 2019-01-06(Sun) 14:00 (local time) (300 minutes)

A - Frog 1

Editorial

Time Limit: 2 sec / Memory Limit: 1024 MiB

Score : 100 points

Problem Statement

There are N stones, numbered $1, 2, \ldots, N$. For each i ($1 \leq i \leq N$), the height of Stone i is h_i.

There is a frog who is initially on Stone 1. He will repeat the following action some number of times to reach Stone N:

- If the frog is currently on Stone i, jump to Stone $i+1$ or Stone $i+2$. Here, a cost of $|h_i - h_j|$ is incurred, where j is the stone to land on.

Find the minimum possible total cost incurred before the frog reaches Stone N.

Constraints

- All values in input are integers.
- $2 \leq N \leq 10^5$
- $1 \leq h_i \leq 10^4$

Input

Input is given from Standard Input in the following format:

$$N$$
$$h_1 \quad h_2 \quad \ldots \quad h_N$$

Output

Print the minimum possible total cost incurred.

Sample Input 1

```
4
10 30 40 20
```

Sample Output 1

```
30
```

If we follow the path $1 \to 2 \to 4$, the total cost incurred would be $|10-30| + |30-20| = 30$.

Sample Input 2

```
2
10 10
```

Sample Output 2

```
0
```

If we follow the path $1 \to 2$, the total cost incurred would be $|10-10| = 0$.

Sample Input 3

```
6
30 10 60 10 60 50
```

Sample Output 3

```
40
```

If we follow the path $1 \to 3 \to 5 \to 6$, the total cost incurred would be $|30-60| + |60-60| + |60-50| = 40$.

Language C++23 (GCC 15.2.0)

Source Code

Open File
Customize
Toggle Editor
Auto Height

* at most 512 KiB

Submit

2026-04-19 (Sun)
18:47:05 -03:00

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